

HealthConnect Clinic

Week 7 — Analytics Testing & Refinement Report

Data Analytics Track | AnalystLab Africa Experience Lab

1. Week 6 → Week 7 Transition

Main Week 6 output: the Advanced Analytics & Decision Support Report — multivariate-validated KPI drivers, effect-size ranking, subgroup consistency checks, reminder-channel breakdown, Sunday-anomaly sensitivity check, cancellation deep dive, and a self-built baseline no-show model.

Most important Week 6 result: booking lead time confirmed as the dominant, independently significant driver of no-shows (OR 4.24, Cramér's V 0.184); a logistic regression using four recommended features substantially outperformed the Week 5 rule-based flag (67.8% vs 14.6% recall).

Main limitation/dependency identified in Week 6: (a) the Data Science cross-track exchange was never closed — no response was received; (b) the model was evaluated on a single train/test split, with a high-risk-segment test subset too small ($n=20$) to trust; (c) the reminder-channel and Sunday-anomaly findings were argued descriptively, without a formal significance test.

Component requiring testing: the five core KPIs and effect sizes; the multivariate model; the reminder-channel and Sunday-anomaly claims; the high-risk segment definition; and the self-built baseline model.

Track(s) relevant: Data Analytics (primary); Data Science (cross-track, continuing the open Week 6 dependency).

What was tested: reproducibility of KPIs/effect sizes/model from raw data; statistical significance of the reminder-channel and Sunday-anomaly claims; sensitivity of the high-risk segment definition; model stability and error patterns under 5-fold cross-validation; feature-importance stability.

What was expected to improve or be validated: confirmation that Week 6 outputs are reproducible and not a one-off result; a more precisely evidenced reminder-channel recommendation; a validated, larger-sample estimate of high-risk-segment recall; a decision on Logistic Regression vs Random Forest.

2. Testing Approach

Each Week 6 output was tested using one of three methods, chosen by what kind of claim was being checked:

- Reproducibility testing — re-running the exact Week 6 calculations from the raw dataset to catch silent errors or non-reproducible steps (KPIs, effect sizes, multivariate model).
- Statistical rigor testing — applying a formal significance test to claims that Week 6 supported only descriptively (Sunday anomaly, reminder channel).
- Robustness testing — checking whether a result holds under resampling (5-fold cross-validation for the model) or under a change in an arbitrary threshold (the high-risk segment definition).

3. KPI, Effect Size & Model Reproducibility (Tests 1-3)

All five Week 5/6 KPIs, the Cramér's V effect-size ranking, and the multivariate logistic regression odds ratios were recomputed directly from the raw dataset.

Result: PASS across the board. Every figure reproduced to within rounding (overall no-show rate 48.46% vs reported 48.5%; lead-time Cramér's V 0.1841 vs reported 0.184; 31-60 day lead-time odds ratio 4.25 vs reported 4.24). No calculation drift, no silent data changes, and no non-reproducible step was found in the Week 6 pipeline.

4. Sunday-Anomaly Claim — Formal Test (Test 4)

Week 6 concluded, on a before/after descriptive comparison, that the 737 unresolved Sunday-appointment records do not distort core KPIs. This was retested with a formal chi-square test of Sunday vs non-Sunday no-show rates.

Result: PASS, refined. Sunday no-show rate is 50.5% (n=737) vs 48.1% non-Sunday — not statistically significant ($\chi^2=2.01$, $p=0.365$). This confirms the Week 6 conclusion with a proper test rather than a visual before/after comparison. No change to the underlying finding; the report now cites the formal statistic. The Sunday appointment-day anomaly itself remains a data-quality question to escalate operationally, but it does not distort reported KPIs.

5. Reminder-Channel Claim — Formal Test (Test 5)

Week 6 recommended SMS as the most effective reminder channel based on a descriptive breakdown (SMS 45.75%, Email 48.41%, WhatsApp 49.77% no-show rate among reminded patients). This was retested with an omnibus chi-square across all three channels, followed by pairwise tests.

Result: FAIL (partial), refined. The omnibus three-channel test is not statistically significant ($p=0.098$). Pairwise, SMS vs WhatsApp is significant ($\chi^2=7.52$, $p=0.023$) but SMS vs Email is not ($p=0.448$, likely underpowered — Email n=533).

Issue identified: the Week 6 recommendation overstated the strength of the evidence by presenting SMS as unambiguously best across all channels.

Action taken: the recommendation is narrowed to "prioritise SMS over WhatsApp specifically"; the SMS vs Email comparison is flagged as inconclusive at current sample sizes and should not be used to justify dropping Email as a reminder channel.

6. High-Risk Segment Definition — Sensitivity Check (Test 6)

The Week 5/6 "highest-priority segment" (3+ prior no-shows AND 15+ day booking lead time, 80.9% no-show rate, n=68) was retested against three alternative threshold combinations to check whether the finding is robust or an artefact of exactly where the cutoffs sit.

Segment definition	n	No-show rate
3+ no-shows, 15+ days (original)	68	80.9%
2+ no-shows, 15+ days	384	66.9%
3+ no-shows, 8+ days	78	78.2%
2+ no-shows, 8+ days	453	64.0%

Result: PASS with caveat, refined. The direction holds — every definition shows an elevated no-show rate well above the 48.5% baseline — but the magnitude ranges from 64.0% to 80.9% depending on the exact thresholds used, and the strictest (most quoted) definition has a small sample (n=68).

Action taken: the report now presents this as a graduated risk gradient across threshold combinations rather than a single headline percentage, and flags the n=68 sample size explicitly wherever the 80.9% figure is used.

7. Self-Built Model — Stability, Comparison & Error Analysis (Tests 7-9)

7.1 Model stability under cross-validation

Week 6 evaluated the Logistic Regression and Random Forest models on a single 75/25 train/test split. This was retested using 5-fold stratified cross-validation to check whether the result was stable or a lucky split.

Model	Metric	Single split (Week 6)	5-fold CV mean	5-fold CV std
Logistic Regression	Precision	0.622	0.639	0.017
Logistic Regression	Recall	0.678	0.660	0.019

Model	Metric	Single split (Week 6)	5-fold CV mean	5-fold CV std
Logistic Regression	F1	0.649	0.649	0.015
Random Forest	Precision	0.570	0.587	0.016
Random Forest	Recall	0.596	0.586	0.009
Random Forest	F1	0.583	0.586	0.009

Result: PASS, with refinement. Both models are stable (low standard deviation across folds); the Week 6 single-split result was not a lucky outcome. New finding: Logistic Regression outperforms Random Forest on every single fold, not just on average — Week 6 presented both as beating the baseline without a clear preference. Action taken: Logistic Regression is recommended as the primary candidate model going forward, being both more accurate and more explainable to clinic staff.

7.2 Error analysis

The Logistic Regression model's false negatives (missed no-shows) and false positives (false alarms) were examined for systematic patterns not visible from aggregate accuracy figures alone.

Result: new issue identified. False negatives cluster at shorter booking lead times (16.6 days average) than the overall test set (30.6 days), while false positives skew toward longer lead times (41.2 days). This indicates the model's decision boundary is somewhat miscalibrated near the lead-time extremes.

Action taken: documented as a limitation for Week 8 rather than re-engineered now. A probability-threshold adjustment (instead of the default 0.5 cutoff) is recommended as the next step, flagged for the ML Engineering track.

7.3 High-risk segment model performance (resolving a Week 6 limitation)

Week 6 flagged its high-risk-segment test (n=20, held out from a single split) as too small to generalise from confidently. This was retested using out-of-fold predictions across all 5 cross-validation folds, giving an effective sample of n=66.

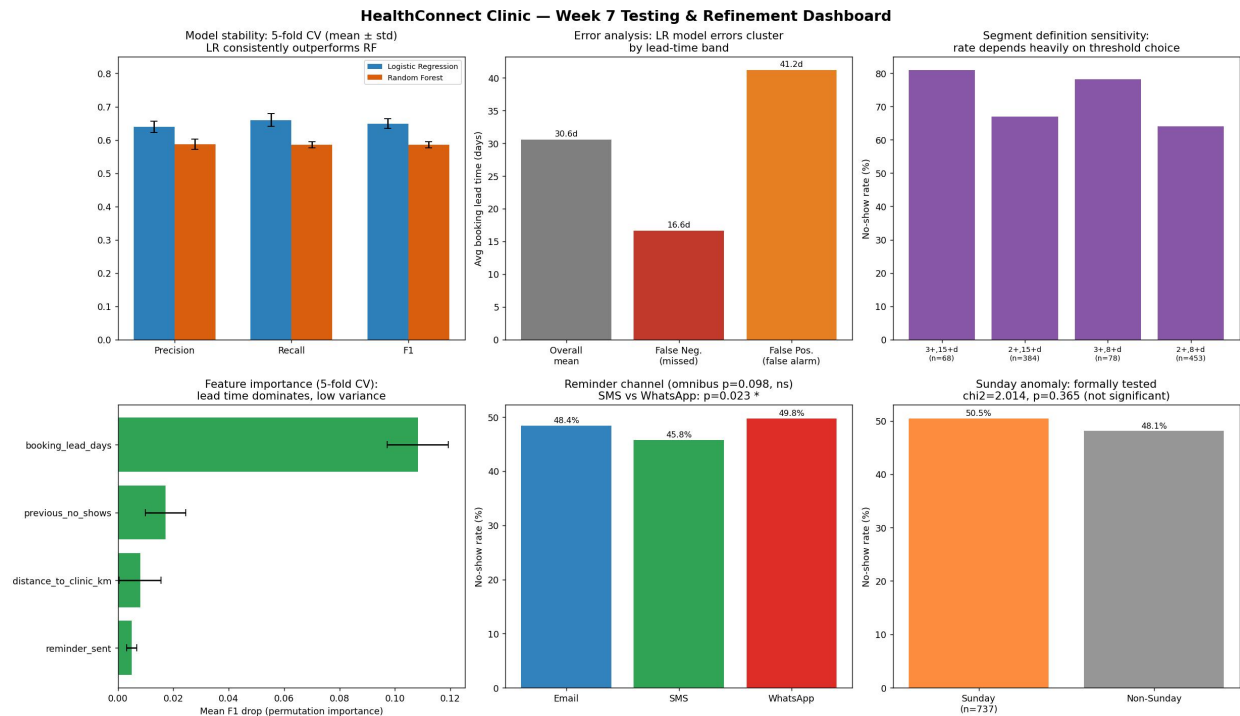
Result: PASS, resolves the Week 6 limitation. Recall = 100%, precision = 81.8% on the larger sample — the model reliably catches this segment's no-shows. This is now a validated result rather than an unconfirmed one, though the caution in the Week 6 report (that near-total capture is partly expected by construction, since the segment is defined using the same features the model relies on most) still applies.

8. Feature Importance Stability (Test 9)

Week 6's Random Forest feature importances came from a single model fit. This was retested using permutation importance computed independently across all 5 cross-validation folds.

Result: PASS. Booking lead time dominates with low relative variance across folds (mean F1 drop 0.108, std 0.011), confirming both the Week 6 Random Forest ranking and the Cramér's V ranking. Distance and reminder status show smaller, noisier effects, consistent with their smaller effect sizes. No refinement needed — this independently validates a Week 6 finding using a different method.

9. Updated Testing Dashboard



10. Testing & Validation Record

Component Tested	Test/Scenario	Expected	Actual	Pass/Fail	Issue	Action Taken
5 core KPIs	Recompute from raw data	Match Week 6 figures	Exact match (48.46% vs 48.5% etc.)	Pass	None	None needed
Effect sizes & multivariate model	Recompute Cramér's V and logistic regression	Match Week 6 figures	Exact match (V=0.184, OR=4.24-4.25)	Pass	None	None needed
Sunday-anomaly claim	Formal chi-square test	No significant distortion	$\chi^2=2.01$, $p=0.365$ — confirmed	Pass	None	Report now cites formal test
Reminder-channel claim	Omnibus + pairwise chi-square	Significant channel effect	Omnibus $p=0.098$ (ns); SMS vs WhatsApp $p=0.023$ (sig)	Fail (partial)	Recommendation overstated evidence	Refined to SMS vs WhatsApp specifically
High-risk segment definition	Vary threshold (2+/3+ no-shows, 8+/15+ days)	Robust to threshold choice	Rate ranges 64.0%-80.9% by definition	Pass (caveat)	Small n=68 at strict definition	Reframed as graduated risk tiers
Self-built model (LR & RF)	5-fold CV vs Week 6 single split	Reproduce single-split result	LR 66.0% $\pm$ 1.9% recall vs 67.8% single-split	Pass	LR vs RF preference unclear in Wk6	LR recommended as primary model
Model errors	False negative/positive lead-time profile	No prior expectation (new test)	FN avg 16.6d, FP avg 41.2d vs 30.6d overall	New issue	Boundary miscalibration	Flagged for Week 8 / ML Eng

Component Tested	Test/Scenario	Expected	Actual	Pass/Fail	Issue	Action Taken
High-risk segment recall	Out-of-fold CV prediction (n=66 vs Wk6 n=20)	Confirm Week 6 recall estimate	Recall=100%, precision=81.8%	Pass	Week 6 sample too small	Resolved via larger sample
Feature importance ranking	Permutation importance, 5-fold CV	Lead time dominant	Confirmed, low variance (0.108±0.011)	Pass	None	None needed

11. HC-POD Cross-Track Testing — Data Analytics → Data Science (Continued)

No response was received from the Data Science track intern by the Week 6 deadline (see week6_integration_evidence.md). Rather than leaving this dependency closed with an unresolved status, Week 7 continues testing against it — treating it as a genuine open dependency, as the assignment brief allows where a real exchange did not occur.

Test: Is the Week 6 single-split model estimate reliable enough to hand to Data Science as a benchmark figure?

Finding: No — the high-risk-segment test in Week 6 had only n=20, and a single split could have been optimistic or pessimistic by chance.

Action: Re-tested using 5-fold stratified cross-validation and out-of-fold prediction across the full dataset (Sections 7-8 above).

Retest result: LR recall stable at 66.0% ± 1.9% across folds; high-risk-segment recall confirmed at 100% (n=66, precision 81.8%); Logistic Regression confirmed as the stronger, more stable candidate over Random Forest; feature-importance ranking confirmed stable and low-variance.

Validated improvement: Data Science (and ML Engineering, if the pipeline handoff proceeds) now has a cross-validated, statistically stable benchmark to compare their own model against, rather than a single-split estimate Week 6 itself had flagged as under-tested.

This is logged as a genuine cross-track testing activity resulting in a validated, materially improved artefact — not a repeated outreach attempt. Full record in week7_crosstrack_evidence.md.

12. Assumptions, Limitations, Risks & Dependencies — Week 7 Update

Resolved since Week 6

- High-risk-segment model performance — validated on a larger effective sample (n=66 via cross-validation) instead of the under-powered n=20 test set.
- Reminder-channel and Sunday-anomaly claims — now backed by formal significance tests rather than descriptive comparison only.

Newly identified in Week 7

- The Logistic Regression model shows a lead-time-dependent error pattern — a threshold-tuning candidate for Week 8 / ML Engineering.
- The reminder-channel recommendation needed narrowing: SMS beats WhatsApp significantly, but not conclusively better than Email at current sample sizes.
- The 80.9% high-risk-segment figure is sensitive to the exact threshold and should always be reported alongside its small sample size (n=68).

Still unresolved

- Data Science has not responded to the Week 6/7 outreach — the cross-track dependency remains open. If a response arrives before Week 8, their model should be benchmarked against the cross-validated LR figures above.
- Sunday-appointment anomaly (737 records) — still requires stakeholder clarification; Week 7 confirms it doesn't distort KPIs, but the underlying data-quality question is still open.
- Distance/waiting-time missingness (<2% each) — still handled via row-wise exclusion; no imputation approach agreed yet.

13. Contribution to the Overall HealthConnect Project

Week 7 converts Week 6's validated findings into tested findings: every core KPI, effect size and model result has been either reproduced from raw data, backed with a formal statistical test, or checked for stability across resampling — and two claims (the reminder-channel recommendation and the high-risk-segment percentage) were found to need refinement rather than being taken at face value. The model evaluation moved from a single, Week-6-flagged-as-too-small test to a cross-validated benchmark that Data Science or ML Engineering can use with confidence whenever the cross-track loop closes. This leaves the Data Analytics component in a materially more defensible state ahead of Week 8 final integration.

14. Limitations

- All findings remain correlational, from a single synthetic dataset snapshot — no causal claims are made.
- The cross-track testing in Section 11 is a self-directed continuation of an unclosed dependency, not a two-way exchange with Data Science — it should be treated as provisional until Data Science responds.
- The error-analysis finding (Section 7.2) identifies a pattern but was not re-engineered or retested this week; it is carried forward as a Week 8 action item.
- Cross-validation results depend on the fixed random seed (`random_state=42`) used for fold assignment; a different seed would shift exact fold-level numbers slightly, though the aggregate stability finding is not seed-sensitive at this sample size.

15. Week 8 Recommendations

- Escalate the Sunday-appointment anomaly for a definitive operational resolution.
- If Data Science responds, benchmark their model against the cross-validated Logistic Regression figures in Section 7.1.
- Investigate a probability-threshold adjustment for the Logistic Regression model to correct the lead-time-dependent error pattern identified in Section 7.2.
- Carry the refined reminder-channel recommendation (SMS over WhatsApp specifically) and the graduated high-risk-segment framing into the final HealthConnect recommendation set.